

Personal homepage of Suk Jin Mun

Design documentation

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1 Project description

Suk Jin Mun is an M.S. Data Science student at Northeastern University who works on data science and machine learning for materials research. The site is his personal homepage, a small website about him and his work. It tells a visitor who he is, where he has done research, what he has published, and how to reach him.

Three groups of readers visit the homepage, each with one short task. Professors who review PhD applications judge research fit, engineers who hire for machine learning roles look for code, and classmates look for contact details. A professor, for instance, opens the homepage from an application and wants to judge research fit in two minutes. Because every reader has only minutes, the design follows one rule, *do not make users think*. Each page answers one question, each link names its target, and each control shows its value.

The site is front end only, which means all of its code runs in the reader's browser. It has three pages at three URLs, listed below. Two pages are built by hand, and the third, the Lab page, is made with an AI coding agent after the first two are done. The Lab page runs a 2D Ising model, a grid of tiny magnets that line up at low temperature and mix at high temperature. All three pages share one header, one footer, and one stylesheet, so a reader learns the layout once. The header holds the navigation bar, called the navbar, with links to all three pages.

- # **Home, index.html.** Built by hand. It opens with the hero, his name and a short about text, followed by three highlight cards, education, and contact.
- # **Research, research.html.** Built by hand. It lists his research interests, six research positions, and publications, and it holds the Bragg peak explorer.
- # **Lab, lab.html.** Made with an AI coding agent. It runs an interactive 2D Ising model in the style of the other two pages.

The research page holds the one original component of the site, the Bragg peak explorer, written from scratch in `js/bragg.js` with no library. The explorer shows the angles at which a cubic crystal reflects light into sharp peaks. A visitor picks a crystal type, a size, and a light source, and the page draws the peaks and lists them in a table. Section 6 specifies the explorer in full. The explorer lets a visitor try working code that Suk Jin wrote, on the page itself.

The site uses no backend, no external API, no jQuery, and no JavaScript framework, as the list below shows with the rest of the setup. GitHub Pages therefore serves it as plain files, so it stays free and keeps working after the course ends.

- # **Markup.** HTML5 with one `h1` per page and standard elements for every control.
- # **Page head.** Every page sets `lang`, `charset`, `viewport`, `description`, `author`, an SVG favicon, and a title.
- # **Style.** `css/style.css` sits on top of Bootstrap 5.3 from the jsDelivr CDN.
- # **Script.** ES6 modules with `type="module"`, with `js/main.js` on every page and `js/bragg.js` and `js/ising.js` on their own pages.
- # **Not used.** No backend, no jQuery, and no React, Vue, Svelte, or Angular.

- # **Folders.** css, js, images, and docs, with the three pages at the root.
- # **Checks.** The class ESLint config, Prettier, and the W3C validator report no errors, and a local server test passes.
- # **Project.** package.json with type module, the MIT license, and small commits.
- # **README.** Author, class link, objective, screenshot, build steps, and a GenAI section.
- # **Hosting.** GitHub Pages, with the site, the design document, the video, and the slides linked from the README.

2 User personas

Each of the three groups of readers from the project description has one persona. A persona is an invented but typical reader with a name, a job, a goal, and a device. Personas matter because each design choice can be checked against one reader's need. Marcus, for instance, takes his first look on a phone, so no page may scroll sideways. Each card gives the persona's goals, frustrations, what the persona looks for, and the device used. Together the three cards set what each page must show first.

Helen, Professor and PhD admissions reader

Goals. She wants to judge research fit in two minutes. She checks publications and supervisors and needs an email address.

Frustrations. Pages that open with animation or long self-description annoy her. So do publications without journal, year, or link.

Looks for. She scans for research interests in plain words, the current lab, publications with links, and email.

Device. She reads on an office desktop with a wide monitor.

Marcus, Machine learning engineer who screens candidates

Goals. He wants to find tools the candidate has built and read their code. He decides in a few minutes whether to pass the profile on.

Frustrations. Pages that scroll sideways on a phone annoy him. So do skill lists with no project behind them, and dead links.

Looks for. He looks for RAPID, a tool Suk Jin built and published, with its code and paper. He also wants one piece of working code on the page.

Device. He takes the first look on a phone and the second on a laptop.

Priya, CS5610 classmate and possible collaborator

Goals. She wants to contact him about the project or shared interests. She also wants to see what he works on now and to try the Lab page.

Frustrations. Contact details hidden in a PDF annoy her. So does navigation that changes from page to page.

Looks for. She looks for email and GitHub on every page, his current work, and the Lab page.

Device. She uses a laptop in class and a phone for quick look-ups.

3 User stories

The personas say what each reader needs, and the user stories turn each need into something the site must do. A user story is one sentence of the form As a ..., I want ..., so that ..., which names the reader, the feature, and the reason. The reason ties each feature to a reader's need. Story 4, for instance, asks for a one-tap menu so that Marcus can read the site on his phone without zooming. Each story adds a short scenario and acceptance criteria, which are checks the finished site must pass. Stories 1 to 3 belong to Helen, 4 to 6 to Marcus, and 7 to 9 to Priya. Together their criteria form a test list for the finished site.

3.1 Helen, professor and PhD admissions reader

Story 1. Research fit on the first screen

As an admissions reader, I want his field and current lab on the first screen, so that I can judge research fit quickly.

Helen opens the homepage from an application. The hero says he works on machine learning for materials research and names his lab at Northeastern. She follows the Current research card to the research page.

Acceptance criteria

- # At 1280 by 800 px the name, role line, and about text show without scrolling.
- # The Current research card names the D2R2 Lab and Prof. Peter Schindler.
- # The Current research card links to the experience list on research.html.

Story 2. Publications with working links

As an admissions reader, I want every publication with its title, journal, year, and a link, so that I can open a paper in one click.

Helen scrolls to Publications. The first paper opens at its DOI, the second leads to his Google Scholar profile, and the third is marked in preparation.

Acceptance criteria

- # Each published paper shows its title, journal, and year.
- # The first paper links to its DOI, and the second links to Google Scholar.
- # The paper in preparation carries no link.

Story 3. Research positions with supervisors

As an admissions reader, I want his research positions in time order with each supervisor, so that I can see how long and with whom he has worked.

Helen reads the experience list from the top. Six entries run from the D2R2 Lab back to a 2018 internship, each with the lab, the supervisor, and the dates.

Acceptance criteria

- # The list has six entries, newest first, each with institution, lab, supervisor, role, and dates.
- # Each entry has one to three short lines in general terms.
- # The Northeastern entry names Prof. Robin Walters as thesis co-advisor.

3.2 Marcus, machine learning engineer

Story 4. A phone layout with no zoom

As an engineer on a phone, I want the site to fit my screen with a one-tap menu, so that I can read it without zooming.

Marcus opens the link on his phone. The navbar shows the name and a menu button, the cards stack in one column, and nothing scrolls sideways.

Acceptance criteria

- # At 375 px no page scrolls sideways, and body text stays at 16 px.
- # The navbar collapses into a toggle button with aria-controls, aria-expanded, and aria-label.
- # Cards and controls stack in one column, and the peak table scrolls inside its own box.

Story 5. RAPID code in two clicks

As an engineer, I want to reach the RAPID code and paper within two clicks, so that I can check a tool he has built.

The RAPID card on the home page says in one sentence what RAPID does and links to its code on GitHub. That evening Marcus opens the repository on his laptop.

Acceptance criteria

- # The RAPID card links to <https://github.com/DataForgeSci/RAPID>.
- # The first publication links to the DOI of the RAPID paper.
- # Every link text names its target, and no link reads click here.

Story 6. Working code on the page

As an engineer, I want to try the Bragg peak explorer, so that I can see working code he wrote.

Marcus drags the size slider to larger values and watches the peaks move left. The table below the canvas follows every change.

Acceptance criteria

- # Every change redraws the canvas and the table at once, with no submit button.
- # The explorer opens on face centered cubic, $a = 4.05 \text{ \AA}$, and Cu $K\alpha$, with five peaks.
- # `js/bragg.js` is an ES6 module written from scratch with no library.

3.3 Priya, classmate and possible collaborator

Story 7. Contact from any page

As a classmate, I want his email, GitHub, and Google Scholar on every page, so that I can reach him in one click.

Priya has a question about his repository. She clicks the email link in the footer, and her mail program opens with his address filled in.

Acceptance criteria

- # The footer of all three pages shows his name, email, GitHub, Google Scholar, and the year.
- # js/main.js writes the current year into the footer.
- # The contact block on the home page repeats the three links with text labels.

Story 8. Current work at a glance

As a possible collaborator, I want a short list of what he works on now, so that I can see where our work overlaps.

Priya works on machine learning. The research interests and the Northeastern entry tell her in a few lines that he does too, and she writes to him the same day.

Acceptance criteria

- # The research interests list is short and uses plain words.
- # The Northeastern entry describes his current projects in general terms.
- # No section shows unpublished results.

Story 9. A Lab page that belongs to the site

As a classmate, I want to open the Lab page from any page and run the Ising model, so that I can play with it in the browser.

Priya clicks Lab in the navbar. The page looks like the other two. She presses Play, lowers the temperature, and watches the grid settle into large patches of one color.

Acceptance criteria

- # Lab appears in the navbar of all three pages, and the current page link has aria-current="page".
- # lab.html uses the same header, footer, stylesheet, and card style as the other pages.
- # The simulation starts only on Play, and the same button then reads Pause.

4 Design mockups

The user stories say what each page must do, and the mockups show where each part of a page sits. Each mockup is a wireframe, a plain drawing of a page that shows its layout but not its final text. Drawing the layout first fixes the order of the parts before any code exists. The phone mockup, for instance, fixes that the three highlight cards stack in one column at 375 px. The mockups show every page at 1280 px for a desktop and the home and research pages at 375 px for a phone. Gray bars stand for text, teal marks links and the active page, dark gray bars name the regions, and gray notes give the Bootstrap classes.

4.1 Site map

The site map is a diagram of the three pages and the links between them. Every page reaches every other page through the navbar. The three highlight cards on the home page, Current research, RAPID, and Publications, link into the research page.

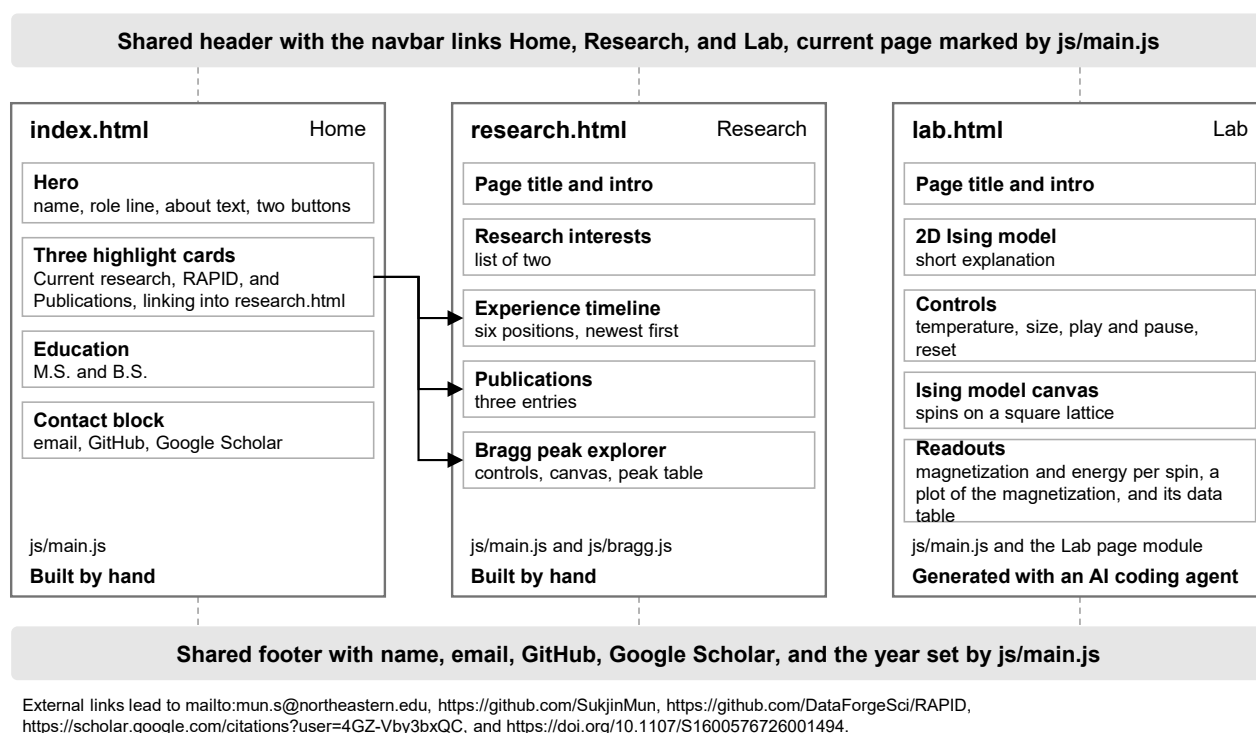


Figure 1. Site map. Arrows show the three highlight cards on the home page linking into the research page.

4.2 Home page on a desktop

The home page is where a reader usually arrives, so its first screen says who Suk Jin is. Helen, for instance, sees his field and current lab without scrolling. The hero has an avatar column and a text column with two buttons, See my research and Email me. The three highlight cards sit on the light gray band below, and the contact block repeats the footer links. A reader can therefore reach the research page or write to him from the first screen.

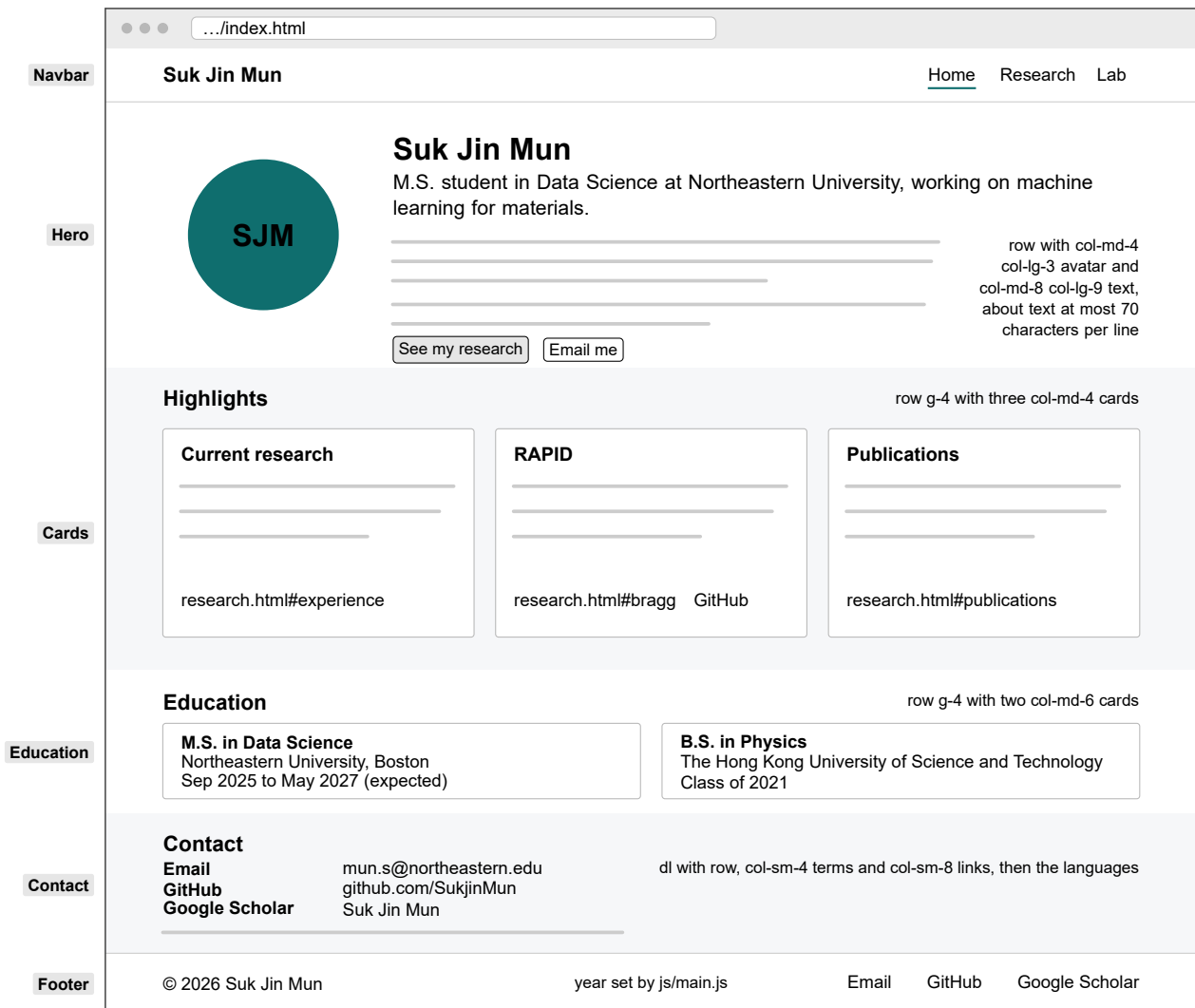


Figure 2. Home page, index.html, at 1280 px.

4.3 Research page on a desktop

The research page holds the record that Helen and Marcus come to check. It lists the research interests, six positions newest first, and three publications, and it ends with the Bragg peak explorer. The explorer sits last, so a reader tries the code after reading the record.

.../research.html

Suk Jin Mun Home Research Lab

Research

Research interests ul with # markers, at most 70 characters per line

Experience vertical list of cards with a teal left border, newest first

- Northeastern University, D2R2 Lab** Sep 2025 to present
Research Assistant, Prof. Peter Schindler
- Institute for Basic Science, Center for Van der Waals Quantum Solids, Pohang** June 2024 to July 2025
Research Assistant, Dr. Sungkyun Choi group
RAPID [RAPID links to its code on GitHub](#)
- Institute for Basic Science, Center for Integrated Nanostructure Physics, Suwon** Feb 2023 to May 2023
Research Assistant, Dr. Sungkyun Choi group
- Max Planck Institute for Solid State Research, Stuttgart** June to Aug 2019 and Jan to Feb 2020
Research Assistant, Thin Film Technology Group, Dr. Gennady Logvenov, Dr. Gideok Kim
- Yonsei University** Sep 2018 to Jan 2019
Research Intern, Prof. Aeree Chung group
- Seoul National University** Mar to May 2018
Research Intern, Prof. Sascha Trippe group

Publications title, journal, year
1. _____ DOI
2. _____ Google Scholar
3. Manuscript in preparation

Bragg peak explorer card with a row g-3 of three col-md-4 controls above the canvas, table below

Crystal type: Face centered cubic Size a: 4.05 Å Light source: Cu Kα, λ = 1.5406 Å

5 peaks between 10° and 90° in 2θ.

Canvas

10° 20° 30° 40° 50° 60° 70° 80° 90°

111 200 220 311 222

hkl	$h^2 + k^2 + l^2$	d (Å)	2θ (°)
111	3	2.3383	38.47
200	4	2.0250	44.72

© 2026 Suk Jin Mun year set by js/main.js Email GitHub Google Scholar

Figure 3. Research page, research.html, at 1280 px. The explorer shows its default five peaks.

4.4 Lab page on a desktop

The Lab page runs the 2D Ising model beside its controls and keeps the header, footer, and band style of the other pages. Each square of the grid is one spin, a tiny magnet that points up or down. Figure 4 is the starting layout for the AI coding agent, and the prompt rounds of Section 7 settle the details.

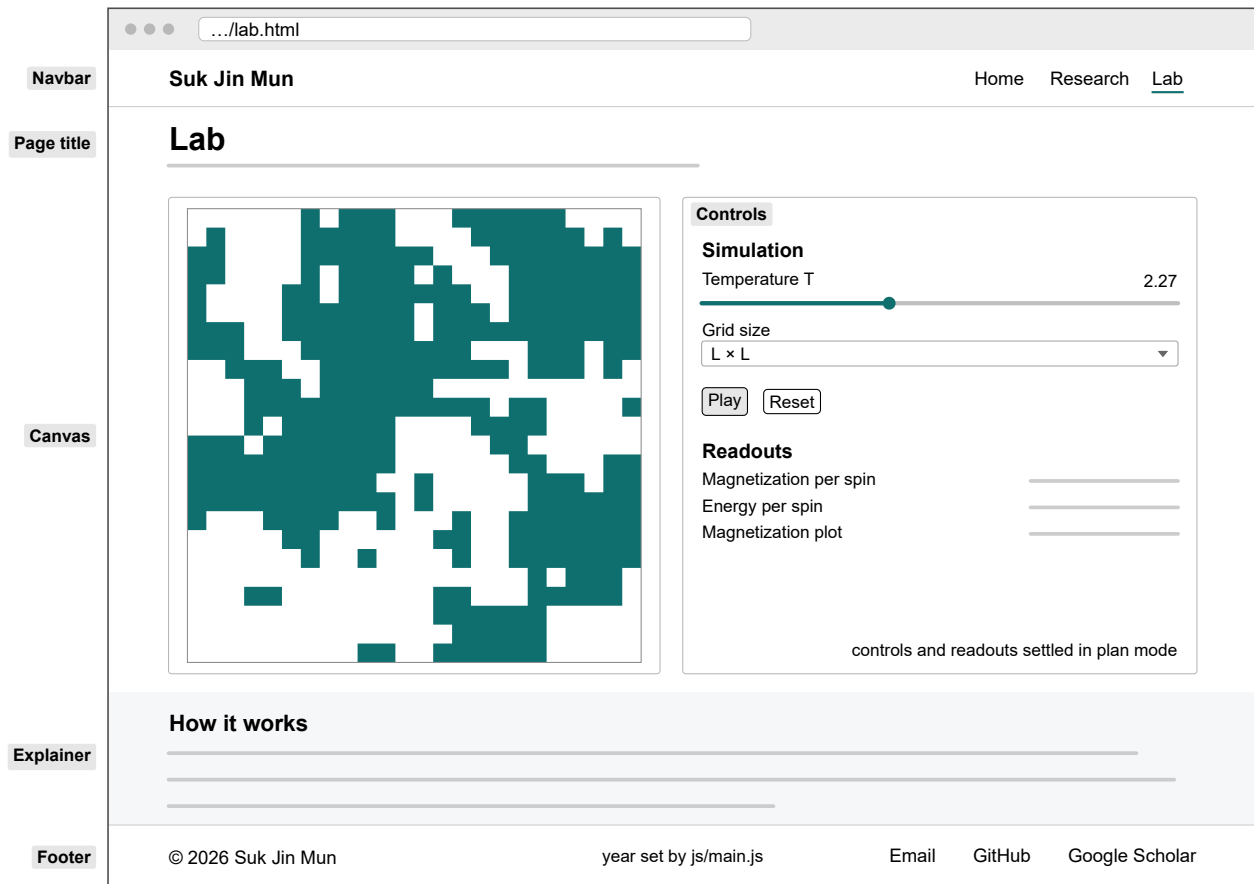


Figure 4. Lab page, lab.html, at 1280 px. Teal squares are up spins.

4.5 Home and research pages on a phone

At 375 px both pages keep the desktop order in one column, so Marcus reads them without zooming.

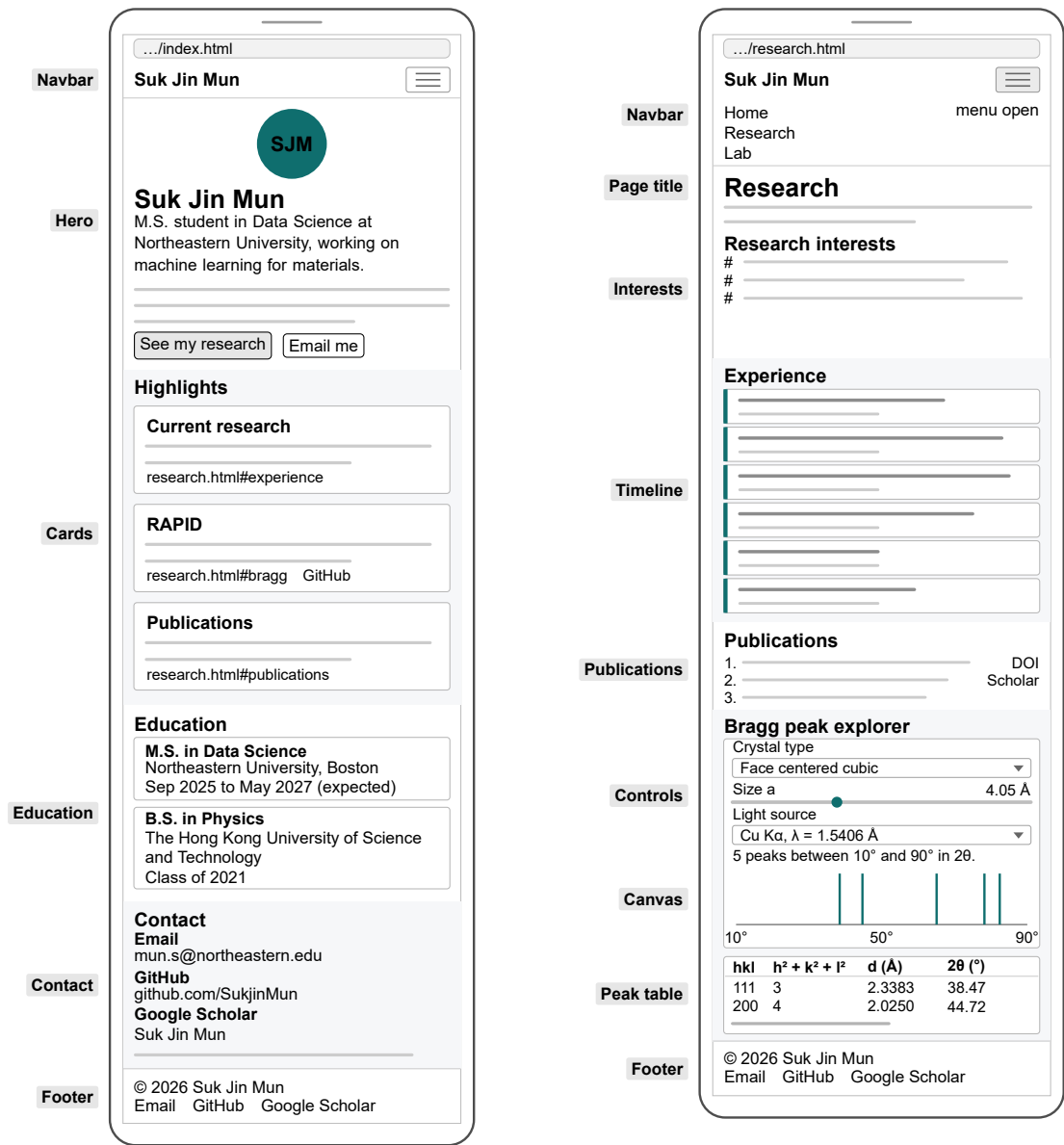


Figure 5. Home page, left, and research page, right, at 375 px. The navbar collapses behind a toggle button, shown open on the right, and rows stack into one column.

5 Style guide

All three pages draw on one style guide, the shared set of colors, type sizes, spacing, and component rules. The style guide matters most for the Lab page, which the AI coding agent must match to the two hand-built pages.

5.1 Palette

The palette, the set of colors on the site, has one accent color, a teal. Text is near black on white or on the light gray band, and teal marks links, the active page, buttons, and the explorer peaks.

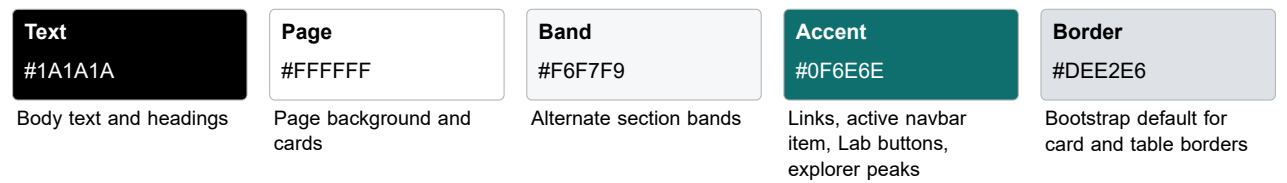


Figure 6. Palette of the site.

Every text pair passes at least level AA of WCAG 2.1, the web accessibility guidelines, for body text.

Table 1. Contrast of every text pair against the WCAG 2.1 levels.

Text on background	Contrast ratio	Level for body text
Text #1A1A1A on page #FFFFFF	17.4 to 1	AAA
Text #1A1A1A on band #F6F7F9	16.2 to 1	AAA
Accent #0F6E6E on page #FFFFFF	6.0 to 1	AA
Accent #0F6E6E on band #F6F7F9	5.6 to 1	AA
White #FFFFFF on accent #0F6E6E, filled buttons	6.0 to 1	AA

5.2 Type

Text uses the Bootstrap system font stack, the fonts already installed on the reader’s device, so no font file loads. Bootstrap scales h1 and h2 down below 1200 px, and the table gives the sizes at 1200 px and wider.

Table 2. Type scale.

Role	Element and class	Size at 1200 px and wider	Weight
Page title, one per page	h1 with class display-5	3 rem, 48 px	300
Section title	h2	2 rem, 32 px	500
Card title	h3 with class h5	1.25 rem, 20 px	500
Role line in the hero	p with class lead	1.25 rem, 20 px	300
Body text	p	1 rem, 16 px, line height 1.5	400
Dates and footer	p	1 rem, 16 px	400

5.3 Layout and lists

Spacing follows the Bootstrap spacer scale, a fixed set of spacing steps. Sections use py-5, card rows use g-4, and paragraphs stop at 70 characters per line. From the md breakpoint, the width of 768 px at which Bootstrap switches layouts, the navbar expands and the cards go three across. Content lists on the site use # as the marker, set with list-style-type: "# " in css/style.css. Navigation lists show no marker.

5.4 Components and rules

The site uses the Bootstrap navbar, grid, cards, buttons, form controls, and a small table, and every page follows the rules below.

- # Each page has one h1, and headings follow in order.
- # Every img has alt text, and every control is a labelled button, select, or input, never a div or span.
- # js/main.js sets aria-current="page" on the current navbar link.
- # Each canvas has role="img" and an aria-label, and a table repeats its data as text.
- # Controls work with the keyboard and show a teal focus outline.
- # Text contrast is at least 4.5 to 1.
- # CSS selects elements by class, with no inline style and no !important.
- # Links and file references use only relative paths.

Together these rules make the hand-built pages and the Lab page look and behave as one site.

6 Bragg peak explorer

The Bragg peak explorer is the original component on the research page, written in js/bragg.js with no library. When light of one wavelength hits a crystal, the planes of atoms reflect it strongly only at certain angles. Each such angle shows up as a sharp peak, called a Bragg peak. The explorer computes these peaks for a cubic crystal, whose atoms sit on a grid of equal cubes. It matters because it is working code a visitor can try, which is what Marcus looks for.

The explorer takes three inputs, each set by a labelled control with a default value.

Table 3. Inputs. Each change redraws the output at once.

Input	Control	Values
Crystal type	select with a label	Simple, body centered, or face centered cubic, default face centered
Size a	range with an output	3.00 to 6.00 Å, step 0.01 Å, default 4.05 Å
Light source	select with a label	Cu K α , 1.5406 Å, default, or Mo K α , 0.7107 Å

Each peak comes from a reflection, a family of parallel atom planes labelled by three whole numbers hkl . The angle 2θ lies between the incoming beam and the reflected beam. The calculation turns the three inputs into sorted peak angles in four steps.

1. Loop over $h \geq k \geq l \geq 0$, skipping 000, up to the largest index that can reach $2\theta = 90^\circ$.
2. Keep the reflections the crystal type allows, as Table 4 lists.
3. Group reflections with the same $N = h^2 + k^2 + l^2$, since they share one peak.
4. Compute $d = a / \sqrt{N}$ and $2\theta = 2 \arcsin(\lambda / 2d)$, keep peaks from 10° to 90° , and sort them.

Table 4. Allowed reflections.

Crystal type	Allowed hkl	First allowed N
Simple cubic	every hkl	1, 2, 3, 4, 5, 6, 8, 9
Body centered cubic	$h + k + l$ even	2, 4, 6, 8, 10, 12, 14, 16
Face centered cubic	h, k, l all even or all odd	3, 4, 8, 11, 12, 16, 19, 20

The canvas draws one teal stick per peak on a 2θ axis from 10° to 90° . The peak table lists hkl , N , d , and 2θ in a scroll box. An aria-live line, which a screen reader announces on every change, states the peak count. With the default inputs the explorer shows the five peaks of Table 5, so a visitor sees a full result before touching any control.

Table 5. Default peaks, face centered cubic with $a = 4.05 \text{ \AA}$ and Cu $K\alpha$.

hkl	N	$d, \text{ \AA}$	$\lambda / 2d$	$2\theta, \text{ degrees}$
111	3	2.3383	0.3294	38.47
200	4	2.0250	0.3804	44.72
220	8	1.4319	0.5380	65.09
311	11	1.2211	0.6308	78.22
222	12	1.1691	0.6589	82.43

7 Plan for the AI generated Lab page

The two hand-built pages come first, and the Lab page follows as the one page made with an AI coding agent. It shows the 2D Ising model, the grid of tiny magnets that line up at low temperature and mix at high temperature. The session starts in plan mode, a setting in which the agent proposes a design and writes no code until the design is agreed. Planning first matters because the agent must match two pages it did not write. Round 4, for instance, checks that the header and footer of lab.html match the other pages. The session opens with the prompt below, which sets the role, the tools, and the style files.

Planned opening prompt

Think as a professional full stack developer with 20 years of experience. Do not implement anything until we agree on the design. I want a new page, lab.html, with an interactive 2D Ising model. Follow the style of the attached index.html, research.html, css/style.css, and js/main.js. Use vanilla HTML, CSS, ES6 modules, and Bootstrap 5.3 only. Ask me questions first.

The agent then works through at least ten rounds, listed in Table 6. Suk Jin reads and tests each round on a local server before the next prompt. The GenAI section of the README records the model and every prompt.

Table 6. Planned prompt rounds for lab.html.

Round	Prompt	Check before the next round
1	Opening prompt in plan mode	The agent asks questions and writes no code
2	Answers on controls, size, and readouts	The plan fits Figure 4
3	A written plan	It lists every file it will touch
4	Approval, then lab.html and its module	Header and footer match the other pages
5	Style fixes	No inline style or !important
6	Simulation check	Patches form at low temperature
7	Controls and animation	Play toggles to Pause, Reset works
8	Accessibility	Labels, keyboard use, visible focus
9	Phone layout	No sideways scroll at 375 px
10	Lint and validator fixes	ESLint, Prettier, and W3C pass

The last round leaves a Lab page that passes ESLint, Prettier, and the W3C validator, the same checks as the hand-built pages.